

LightSeizureNet: A Lightweight Deep Learning Model for Real-Time Epileptic Seizure Detection

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Abstract—The monitoring of epilepsy patients in non-hospital environment is highly desirable, where ultra-low power wearable seizure detection devices are essential in such a system. The state-of-the-art epileptic seizure detection algorithms targeting such devices either rely on manual feature extractions, which can be biased due to the experience of experts, or deep neural networks, which suffer from high computation complexity. In this paper, we propose a lightweight deep learning model, LightSeizureNet (LSN), for real-time epileptic seizure detection based on raw EEG data in ultra-low power wearable seizure detection devices. The proposed LSN model includes a patient-independent version and a patient-specific version, both of which avoids manual feature extractions and high computation complexity, while maintaining good classification accuracy. Dilated one-dimensional (1D) convolution, global average pooling, and kernel-wise pruning are adopted to compress the LSN model. The proposed models are evaluated on the CHB-MIT scalp EEG database. The patient-independent LSN model achieves 97.09% accuracy with 6.2M MACs, while the patient-specific LSN model achieves 99.77% accuracy with 3.7 M MACs, which are competitive compared to the state of the art in terms of accuracy and complexity. Furthermore, the proposed model is highly interpretable, which is missing in many previous works. By using a uniform approach to explore the interpretability of the proposed model, fine-grained information such as the activated brain region and the frequency of brainwave during seizures is obtained for clinical diagnosis.

Index Terms—Deep neural network, epilepsy, EEG, feature extraction, interpretability, lightweight model, structure pruning.

I. INTRODUCTION

EPILEPSY is one of the most common neurological diseases, affecting more than 65 million people around the world [1]. Epilepsy is characterized predominantly by recurrent and unpredictable interruptions of normal brain function, called epileptic seizures. Epileptic seizure is defined as a transient

occurrence of symptoms due to abnormal excessive or synchronous neuronal activity in the brain [2]. As a chronic disease, epilepsy itself is not fatal. However, patients lose consciousness during epileptic seizures, which can put the patients in disastrous situations.

Electroencephalogram (EEG) signal is commonly used by experts in the diagnosis of neurological disorders, including epileptic seizures, which are characterized by the frequent occurrence of spikes in EEG signals. The scalp EEG is a non-invasive measure of the electrical potentials generated by the activity of neurons in the brain. The scalp EEG can be collected with electrodes on the scalp. Continuous monitoring of brain electrophysiological activity is necessary for epilepsy patients. Many of the epilepsy patients have to go to specialized hospitals to receive continuous monitoring of their brainwave (EEG) signals. The long-term EEG visual inspection is extremely time-consuming and laborious for physicians to identify characteristic patterns of seizures from multiple channels [3]. In addition to the application of long-term EEG visual inspection, close-loop EEG systems have been developed to suppress the seizure through neural stimulation [55], [56]. The close-loop brain stimulation systems require prompt and accurate response for seizure and can provide stimulation to prevent the onset or spread of seizure activity in the early phase of a seizure. From visual inspection of long-term EEG to close-loop brain stimulation systems, the continuous monitoring of EEG signals in epilepsy patients is essential. Once the epilepsy patients are away from the hospital, they receive extra care in low-acute setting. An ultra-low power wearable device for automatic epileptic seizure detection in non-hospital environment is highly desirable, where lightweight detection algorithms with high accuracy are essential.

EEG-based epileptic seizure detection is typically performed with machine learning techniques. Conventional machine learning methods for EEG-based epileptic seizure detection includes three main stages: preprocessing, feature extraction, and classification. In particular, feature extraction is the key step, which has significant impact on the performance of classification. Many studies have achieved promising results with manual feature extraction. However, those handcrafted features can be biased due to the experience of experts. Furthermore, the features of the EEG signals vary among different patients. Even for the same patient, the EEG signals could change substantially when the patient is at different modes, status (awake or asleep), or ages [4]. When those algorithms are implemented on wearable devices, the feature extraction schemes cannot be changed any more, which limits the applicability of the devices. Therefore, seizure detection algorithms that are free of manual feature extraction are preferable.

With the development of deep learning technology, deep neural network shows its unique ability to learn features from raw

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EEG data. Recently, deep learning methods have outperformed conventional machine learning methods by learning more distinct and robust features from raw EEG data compared to the handcrafted features [5]. Thus, deep seizure detection model is necessary to directly learn features from raw EEG data, so that the pitfalls with manual feature engineering can be avoided.

Despite the promising results reported by previous deep learning methods on EEG-based epileptic seizure detection, there are still several challenges to address. First, typical deep learning models contain a large number of parameters and computations (e.g., VGG16 contains 138M parameters and 15.6G multiply-accumulate operations (MACs) [6]), which puts forward high requirements for the computation and storage capacity of wearable devices. The demonstrated models in many previous studies are not optimized specifically for EEG signal processing. Some works even directly employ the classical models that have been successful in the field of computer vision, such as ResNet and VGG [7]. Such models cannot extract features from EEG data effectively. To obtain high classification accuracy, the scale of those models is very large. To obtain a concise and effective model, we must build the seizure detection models according to the characteristics of EEG signals.

Another issue with current EEG-based epileptic seizure detection approaches, for biomedicine and healthcare, is the interpretability of the deep learning models. Clinically, seizures are classified as focal, generalized, or unknown [8]. Focal-onset seizures and generalized-onset seizures are defined according to the original point and their subsequent development. A focal onset may correspond to a treatable focal pathology, which needs to be clarified through detailed EEG monitoring [9]. However, previous studies focus more on classifying the seizure status but overlook collecting fine-grained information such as the activated brain regions and signal frequency band. In this study, we investigate the interpretability of the proposed model from two aspects: EEG channels and kernels. The information obtained would be valuable for clinical diagnosis.

To handle the challenges mentioned above, we propose a lightweight and interpretable deep learning model for real time epileptic seizure detection, LightSeizureNet (LSN), which can directly learn features from raw EEG data. The proposed LSN model can not only classify the seizure status, but also identify the activated EEG channels and its corresponding frequency band, which is ignored by most previous studies.

The main contributions of this work are highlighted as follows:

- We propose a lightweight and interpretable neural network model for seizure detection, which can effectively extract features from raw EEG signals.
- We adopt dilated 1D convolution, global average pooling, and structure pruning for model compression. Particularly, we propose a specialized structure pruning approach to automatically capture the importance of convolution kernels.
- We present a uniform approach to explore the interpretability of deep seizure detection model for better analyzing the information of seizure from the perspective of neural network model.
- We demonstrate by experiments that the proposed LSN model achieves a competitive performance compared with the existing seizure detection methods, with a small number of parameters and computations.

This paper is organized as follows. The existing EEG-based epileptic seizure detection methods are reviewed in Section II. The lightweight deep neural network model LightSeizureNet is proposed in Section III. Experimental results are presented in Section IV to evaluate the performance and analyze the interpretability of the proposed model. The paper is discussed in Section V and concluded in Section VI.

II. RELATED WORKS

EEG-based epileptic seizure detection can be divided into two categories: conventional machine learning methods and deep learning approaches. Below, we briefly review the most relevant studies.

A. Conventional Machine Learning Methods

Being the key for conventional machine learning methods, feature extraction has the greatest impact on classification performance. In general, most of the existing works achieve promising performance with expert handcrafted features, which can be broadly classified into four categories: time domain, frequency domain, time-frequency domain, and spatio-temporal domain features.

The statistical distributions of EEG signals are different between seizures and normal periods. Therefore, some works employ features in the time domain, such as Approximate Entropy (ApEn) [10], Sample Entropy (SamEn) [11], and other statistical features [12]. However, such statistical features usually require a large number of calculations. For example, the time complexity of ApEn and SamEn are both $O(n^2)$, where n is the length of time series. Those feature extraction algorithms are therefore not suitable for online applications with long data. Furthermore, the complex data scheduling of these feature extraction algorithms is challenging for hardware implementation.

The frequency characteristics of the EEG signal are changed during the epileptic seizures. Therefore, frequency features are appropriate candidates for seizure detection. To obtain frequency-domain based features, the popular techniques include FFT based Welch spectral analysis [13] and Discrete Cosine Transform (DCT) [14]. Due to the non-stationary nature of EEG signal, time-frequency analysis methods are also employed, such as Short Time Fourier Transform (STFT) [15], Wavelet Transform (WT) [16], and Empirical Wavelet Transform (EWT) [17]. Typically, researchers transform EEG signals to frequency domain or time-frequency domain firstly, and then extract well-designed handcrafted features. Therefore, additional hardware is required for these transformations, which causes burden to the wearable devices.

To capture the spatio-temporal propagation of a seizure for seizure detection, the spatio-temporal domain features such as phase-locking value [18], mean phase coherence [19], and dynamical similarity [20], are widely used. However, the classification accuracy of the seizure detection methods using these features is not high. Therefore, spatio-temporal features have received less attention in recent years.

Generally, researchers tend to divide EEG signals into five frequency sub-bands of clinical interest, i.e., delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (13–30 Hz), and gamma (30–128 Hz), and extract features from them. The dimension of features increases rapidly with the increase of the number of EEG channels and frequency sub-bands. Thus, feature selection

becomes a great challenge to conventional machine learning methods. To avoid dimension curse, some techniques, such as PCA, ICA [21], and Laplacian Eigenmaps [22], are employed. However, the time complexity of PCA is $O(n^2)$. The time complexity of ICA is $O(kn^2)$, where k denotes the number of data samples [23]. The time complexity of Laplacian Eigenmaps is $O(n^3)$ [24]. The methods mentioned above are time consuming and rely heavily on human expertise. Implementing some of those algorithms on hardware is even more complicated compared to deep learning networks.

B. Deep Learning Methods

Recently, deep learning methods show great potential in automatic seizure detection. Some works combine conventional feature extractions and deep neural networks to detect seizures [25], [48]. Ansari et al. [25] extract 55 types of features including frequency domain, time domain, and entropy-based features. A deep Convolutional Neural Network (CNN) and random forest are employed for feature selecting and classification, respectively. Although the CNN only contains 3300 parameters, a wide variety of features still make the seizure detection approach complicated. Ke et al. [48] measure the global maximal information coefficient (MIC) of all EEG data channels as features and adopt a lightweight VGGNet as classifier. The lightweight VGGNet only contains 50k parameters. However, even if an approximate algorithm is used, the computational complexity of MIC is still $O(n^{2.4})$ [26]. Considering the length of EEG segments in [48] is 2048, the total amount of calculation with this method exceeds 1G MACs. Those lightweight neural network models combined with well-designed features can obtain good classification accuracy. However, the drawbacks of feature extractions still exist in those methods.

Some other works tend to remove the feature extraction stage due to its inherent defects. They adopt deep learning techniques to automatically extract features from raw EEG data. [27] and [28] employ CNNs for automatic seizure detection. Firstly, multi-channel EEG signals are concatenated into 2D images. Then, 2D convolution kernels (3×4 [27], 4×5 [28]) are utilized to extract features. Such convolution kernels are not fully suitable for EEG signals since they only have a narrow receptive field in the time dimension.

Recurrent neural network (RNN) is also used for automatic seizure detection. [29] extracts time-frequency features from EEG signals by discrete wavelet transform and employs a Long Short-Term Memory network (LSTM) for classification. [50] combines CNN with RNN for seizure detection. The CNN is used for extracting features, which are then fed into a Bi-directional LSTM (Bi-LSTM) to perform the classification. Bi-LSTM belongs to RNN, which can extract the time-domain information in EEG signals effectively. The inherent sequential nature of recurrent models precludes parallelization [30]. The recurrent models also suffer from gradient vanishing issue, which becomes critical at long sequence lengths.

Three-dimensional convolutional neural network (3D CNN) is adopted in [31] for seizure detection. 3D CNN extends one more dimension based on the classical two-dimensional (2D) CNN, which makes it possible to extract temporal and spatial information simultaneously. However, 3D CNN greatly increases the amount of calculations also due to the one more dimension.

The application of deep learning methods with deep models to EEG-based epileptic seizure detection provides satisfactory

classification performance. However, increased parameter set of deep learning model brings great challenges to the computing and storage capacity of wearable devices. Some works propose low power implementation of seizure detectors with a few electrodes. In [57], [58], a CNN based seizure detector SeizureNet is implemented on an implantable ultra-low power microcontroller. In [59], three seizure detectors are proposed that runs CNN, RNN, and random forest models, respectively, on ultra-low power microcontrollers. These models have small scale and low energy consumption, yet in trade of the classification performance. To address the high complexity issue with deep learning while maintaining competitive classification performance compared to the state of the art, we propose a lightweight neural network model, which can detect epileptic seizures efficiently. We also explore the interpretability of the proposed model to identify the activated brain regions and signal frequency band during seizures, which is often overlooked by previous studies.

III. METHODOLOGY

In this section, we propose a lightweight deep learning model, LightSeizureNet (LSN), for automatic real-time epileptic seizure detection running on ultra-low power wearable devices in non-hospital environment. We focus on reducing the scale of the proposed seizure detection model while maintaining high classification accuracy. The proposed LSN model can directly extract high dimensional features from raw EEG signals and classify the seizure states of the samples. Furthermore, the information obtained by the proposed model can provide fine-grained information such as the activated brain regions and signal frequency band for clinical diagnosis.

The overall architecture of proposed lightweight model LSN is shown in Fig. 1. The proposed model has a one-dimensional CNN (1D-CNN) architecture, which contains seven convolution layers. The first layer of the model uses dilated 1D convolution. The depth-wise 1D convolution and standard 1D convolution are used alternately for the following six convolution layers. All the seven convolution layers are followed by a ReLU function. The EEG sequences of each EEG channel are convoluted separately. Following the seven convolution layers, the obtained activation maps are averaged through a global average pooling (GAP) layer. Then, the activation maps of each EEG channel are flattened and sent into a linear layer. Finally, a softmax layer is adopted to generate the probability of seizure. Meanwhile, we can identify the activated brain region through the activation maps of each EEG channel as illustrate in Fig. 1, which is to be explained in details in Section IV-E.

To obtain a lightweight model, dilated convolutions, global average pooling (GAP), and kernel-wise pruning are employed for model compression, which are discussed in details in Sections III-A, III-B, and III-C, respectively.

A. Dilated 1D Convolution

The raw EEG signal contains low-frequency information with long time periods and high-frequency information with short time periods [32]. 1D convolution can effectively extract features from time series such as EEG signals. Specifically, 1D convolution includes intra-channel convolution and inter-channel convolution operations. Intra-channel convolution can extract time-frequency information, while inter-channel convolution

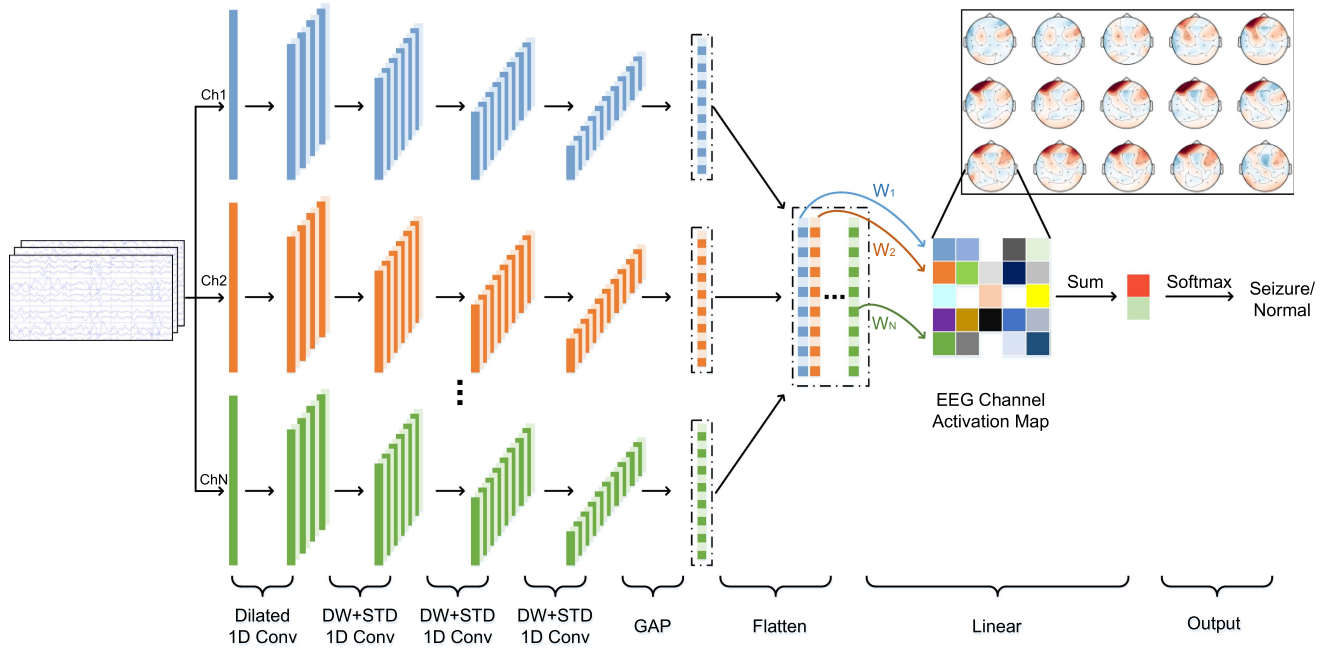


Fig. 1. The overall architecture of the proposed LightSeizureNet (LSN) model. DW: depth-wise. STD: standard. GAP: global average pooling. One EEG window produces one EEG channel activation map.

can extract spatial information from multiple EEG channels. The model with 1D convolution combines the information in time-frequency domain and spatial domain, thereby can accurately classify the seizure states. Meanwhile, due to the reduced dimension, the number of parameters of 1D convolution is significantly less than the 2D and 3D convolutions.

Since 1D convolution operation is similar to the filtering operation, we can analyze the convolution kernel of the model from the perspective of digital filter. Assume the EEG signals are denoted by $\chi = \{X_i \in \mathbb{R}^{C \times T}\}$, in which each EEG sample X_i has T time series and C channels. Specifically, for a given C -channel EEG sample x , the activation of the k -th feature map is obtained by

$$h^k = f\left(\sum_{c=1}^C x_c * W_c^k + b^k\right), \quad (1)$$

where $*$ denotes the 1D convolution, W_c^k denotes the filter of the k -th feature map in the c -th channel, and b^k denotes the bias vector for the k -th feature map.

At the first layer of the proposed model, we remove the bias term and the inter-channel convolution operations. Only intra-channel convolution operations are kept. Therefore, for the n -th EEG channel, (1) degenerates to

$$h_n^k(p) = x_n * W_n^k = \sum_{s+t=p} x_n(s) W_n^k(t), \quad (2)$$

where t denotes the width of the filter. The formula is consistent with the convolution operation of $(t-1)$ order FIR filter, i.e., the EEG signals are filtered by a $(t-1)$ order FIR filter. Therefore, we can explore the interpretability of the model by analyzing these FIR filters. The detailed discussion is in Section IV-E.

Stacked dilated convolutions are employed to obtain large receptive fields with fewer layers while preserving resolution

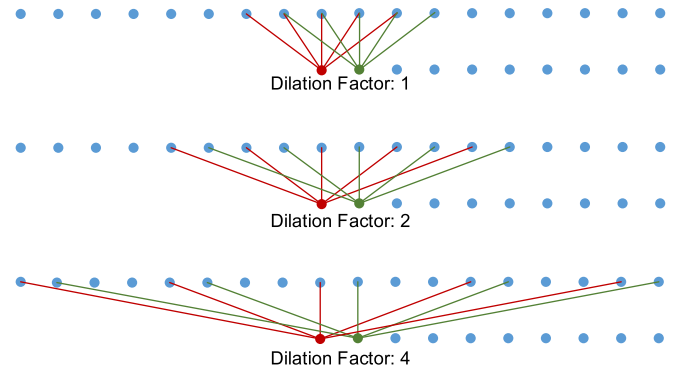


Fig. 2. Illustrations of dilated convolution with different dilation factors.

as well as computational efficiency [33], [34]. In the proposed LSN model, we use short convolution kernels (i.e., 1×5). Furthermore, multi-view 1D dilated convolutions are employed with different dilation factors to improve the feature extraction ability of 1D convolution with short kernels. Formally, let l be a dilation factor and $*_l$ be an l -dilated convolution. l -dilated convolution is defined as

$$x_n *_l W_n^k = \sum_{s+lt=p} x_n(s) W_n^k(t). \quad (3)$$

By adjusting the dilation factor, we can expand the scope of convolution kernel as shown in Fig. 2. Besides analyzing the receptive field of convolution kernel, we also analyze the dilated convolution from the perspective of digital filtering. Dilated convolution can be decomposed into two operations: down sampling and conventional convolution. For $i \in (0, l-1)$, a

l -dilated convolution can also be derived as

$$x_n * l W_n^k = \sum_{s+t=p} x_{ds_n^i}(s) W_n^k(t), \quad (4)$$

where $x_{ds_n^i}(s) = x_n(ls + i)$ denotes the down-sampling of x_n . Specifically, l -dilated convolution is equivalent to down-sampling the signal x with l as the sample rate firstly, which is convoluted by W_n^k afterwards. Therefore, as the dilation factor increases, the bandwidth of the corresponding EEG signals is reduced (i.e., the EEG signal is down-sampled). Specifically, a dilation factor of 2 is equivalent to two times down-sampling of the EEG signals. In the proposed LSN model, we make dilation factor $l \in (1, 2, 4, 8, 16, 32)$. When the sample rate of raw EEG signals is 256 Hz and l is $(1, 2, 4, 8, 16, 32)$, the frequency band of the filtered EEG signals is $(0, 128)$, $(0, 64)$, $(0, 32)$, $(0, 16)$, $(0, 8)$, $(0, 4)$, respectively. The choice for dilation factor is mainly inspired from dominant frequency band of seizure wave (i.e., 0.5–29 Hz) [35].

In summary, we mainly use 1D convolution with short kernels in the model as a trade-off between classification accuracy and computational complexity. To broaden the receptive fields of the kernels and explore the interpretability of the model, we employ dilated 1D convolution with different dilation factor at the first layer.

B. Global Average Pooling (GAP)

In many neural network models, the fully-connected layer typically contains the majority number of parameters. Furthermore, due to the unique fully-connected structure, the weights and activations cannot be well reused. The fully-connected layers consume high memory footprint and bandwidth [36]. We perform global average pooling (GAP) [37] on the convolutional feature maps to avoid the use of fully-connected layers. Therefore, we can minimize the number of parameters while maintaining high classification accuracy. In global average pooling, no parameters need to be trained and optimized. The average values of feature maps of each channel are calculated, and then are directly sent to the softmax layer.

By taking advantage of the simple connectivity structure of global average pooling, we can easily identify the important EEG channels by comparing the weighted output of the activation map of each channel. We propose a technique, which is inspired by [38], to obtain the EEG Channel Activation Maps (ECAMs) that reflect the importance of each EEG channel. The details will be well discussed in Section IV-E.

C. Kernel-wise Pruning

Network pruning is a promising technique for network compression, which can effectively reduce the high computational complexity and frequent memory access caused by deep neural network algorithms. Several studies, such as [40], show that deep neural networks have little or no performance loss even when a large portion of weights are set to zero. Pruning techniques can be broadly categorized into structured or unstructured pruning depending on the pruning granularity. Unstructured pruning is more flexible without many constraints on location and granularity compared to structured pruning. However, the irregular sparsity introduced by unstructured pruning is difficult to be exploited for efficient computation and storage. In contrast, structured pruning constrains that non-zero parameters can only

Algorithm 1: The Proposed Pruning Approach.

Input: Training dataset, X

Output: Optimized model, F

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1:  $W, \alpha \leftarrow$  Initialize network parameters
2: for  $i = 1 \rightarrow Epochs$  do
3:   for  $j = 1 \rightarrow Batches$  do
4:      $\hat{y}_s \leftarrow F(x, W, \alpha)$ 
5:      $L \leftarrow \hat{y}_s, y_s$ 
6:     Minimize  $\frac{1}{N} \sum L(\hat{y}_s, y_s) + R(\alpha)$ 
7:     Update  $W, \alpha$ 
8:   end for
9:   if  $i \% E_{prune} = 0$  then
10:    for  $k = 1 \rightarrow Layers$  do
11:      Find  $\alpha_{min}$  in Layer  $k$ 
12:      if  $\alpha_{min} < \varepsilon$  then
13:        Prune( $\alpha_{min}$ )
14:      end if
15:    end for
16:  end if
17: end for
18: return  $F$ 

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appear at well-defined locations, which enables processing elements to easily exploit computational savings [41]. Therefore, we attempt to minimize the scale of the model by the hardware-friendly structured pruning.

We propose a structured pruning approach with kernel-wise competition. The proposed approach automatically renders retaining of the most important kernels through competition mechanism, while pruning unimportant and irrelevant kernels.

The output of convolutional layers of interest is scaled by a learnt weight α . The dimension of the vector α is consistent with the number of convolutional kernels. To obtain a sparse weight α , we add sparsity constraint on α . Formally, the predicted seizure state $\hat{y}_s$ is derived as

$$\hat{y}_s = F(x, W, \alpha), \quad (5)$$

where $F(\cdot)$ is the function of proposed model. The objective function of our proposed method can be formulated as

$$\min_{w, \alpha} \frac{1}{N} \sum L(y_s, \hat{y}_s) + R(\alpha), \quad (6)$$

where $L(\cdot)$ is a loss function. $R(\cdot)$ is a sparsity regularization for α . In this work, we choose l_2 -norm as the sparsity constraint, because its gradient can be easily derived.

Due to the introduction of sparsity regularization, some weights degrade with the progress of training, while the other relatively important weights are retained to ensure the functionality of the model. In other words, the most important weights are selected through the competition mechanism.

In the proposed pruning algorithm, we introduce two hyperparameters: E_{prune} and ε . E_{prune} denotes the stride of pruning. ε denotes the threshold for pruning. In the training process, the pruning operations are performed at the stride of E_{prune} epochs.

In the pruning process, the absolute value of the weight $|\alpha|$ is calculated. The minimum weight $|\alpha|$ in each layer is selected as a pruning candidate. When the absolute value of the minimum weight $|\alpha|$ in each layer is smaller than the threshold ε , $|\alpha|$ is pruned (set to zero). In each epoch, only the smallest $|\alpha|$ and its corresponding kernel in each layer is pruned. We apply the strategy to ensure that the pruning operation does not have significant impact on the model. According to the proposed pruning algorithm, the corresponding convolution kernels of retained weight α are considered important. Meanwhile, $|\alpha|$ represents the importance of its corresponding convolution kernels. The proposed pruning approach is presented in Algorithm 1. The time complexity of the proposed pruning approach is $O(lm)$, where l is the number of layers and m is the number of kernels. After the training process, the weights of each kernel are multiplied by $|\alpha|$ to obtain the final model. Therefore, the pruning approach has no computing overhead in the inference process.

IV. EXPERIMENT AND RESULTS

A. Dataset

The EEG dataset used in this paper is the CHB-MIT dataset collected from the Children's Hospital Boston [42], [60]. The CHB-MIT dataset consists of EEG recordings from 23 pediatric subjects with intractable seizures, including 5 males and 18 females aging from 2 to 22 years old. In the CHB-MIT dataset, all EEG signals are sampled at 256 Hz with 16-bit resolution. The EEG signals are collected from EEG electrodes placed according to the International 10–20 system. Overall, the dataset contains ~ 983 hours of EEG recordings, which includes 198 seizure events with a total duration of 3 hours 15 minutes.

B. Pre-Processing

The majority of recordings in the CHB-MIT-EEG dataset has 23 channels, while other recordings miss or repeat some channels. In this study, the missing and duplicate channels (T8-P8 and P7-T7) are removed. We use the following 21 channels (based on the international 10–20 system): FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, T7-FT9, FT9-FT10, and FT10-T8.

To remove bias and attenuate power line disturbance, the raw EEG signals of each EEG channel are digitally filtered using a 20th-order Butterworth bandpass filter with the lower and higher cut-off frequency of 0.5 and 45 Hz, respectively. The filtered EEG data are then split into windows of 2 seconds with 50% overlaps. Specifically, each EEG sample contains EEG data from 21 channels with a window size of 512 (i.e., 2 seconds) and an overlap rate of 50%. Since the onset of seizures only accounts for a small proportion of EEG recordings, the obtained dataset is highly skewed, i.e., the number of positive samples is much less than the number of negative samples, which is not conducive to the training of neural network. In general, data augmentation and re-sampling are used to solve this issue. Data augmentation, such as Data splicing [43], Generative Adversarial Networks (GAN) [44], and other strategies [45], all face a challenge in generating authentic artificial EEG samples. These approaches may introduce artificial noise. Therefore, we obtain a balanced dataset by re-sampling. In each epoch of training, we randomly select the same number of negative samples as all positive samples in the training set, and mix them up randomly. Therefore,

TABLE I
THE STRUCTURE OF THE PROPOSED LSN MODEL

Layer Name	Type	Output size	Kernel size / Stride
Input	-	(21, 512)	-
Conv1	Dilation 1D [#]	(21, 12, 512)	2×1×5 / 1
Conv2	Depth-wise 1D	(21, 12, 256)	1×1×5 / 2
Conv3	Standard 1D	(21, 24, 128)	24×12×5 / 2
MaxPooling	-	(21, 24, 128)	1×3 / 1
Conv4	Depth-wise 1D	(21, 24, 64)	1×1×5 / 2
Conv5	Standard 1D	(21, 32, 64)	32×24×5 / 1
MaxPooling	-	(21, 32, 64)	1×3 / 1
Conv6	Depth-wise 1D	(21, 32, 64)	1×1×5 / 2
Conv7	Standard 1D	(21, 48, 32)	48×32×5 / 1
MaxPooling	-	(21, 48, 32)	1×3 / 1
GAP	-	(21, 48)	1×1×32
Linear	-	2	2×21×48

[#]: 6 dilation factors (1, 2, 4, 8, 16, 32).

we obtain a balanced training subset for each epoch. Note that the re-sampling strategy is only adopted in the training process, but not in the evaluation.

C. Training Details and Evaluation

The model is implemented using Python 3.6.5 and PyTorch 1.4.0. The detailed architecture parameters of our proposed lightweight model LSN is listed in Table I. We employ CrossEntropyLoss as the loss function. We adopt the Adam optimizer with learning rate of $5e-4$. The proposed model is trained for 2000 epochs. A dropout rate of 0.4 is added to all the layers to prevent overfitting. The pruning stride E_{prune} is 50. The pruning threshold ε is $1e-2$. The parameters are set based on empirical hyper-parameter tuning.

To evaluate the proposed model in a comprehensive scale, we train a patient-independent model and a patient-specific model. To obtain patient-independent model, the model is trained (and evaluated) for all subjects. Alternatively, to obtain patient-specific model, the model is trained (and evaluated) for each individual patient. We perform 10-fold cross-validation, which is separated based on the number of EEG windows, and report the average performance. The evaluation metrics consist of sensitivity, specificity, and accuracy.

D. Results and Comparisons

To validate the performance of the proposed model and pruning approach, we perform training on the patient-independent model with and without the proposed kernel-wise pruning approach. The model trained without pruning is considered as the baseline. The classification accuracy and loss of the baseline and the pruned patient-independent model for all training epochs are shown in Fig. 3. Note that, the pruned model only shows negligible performance drop compared with the

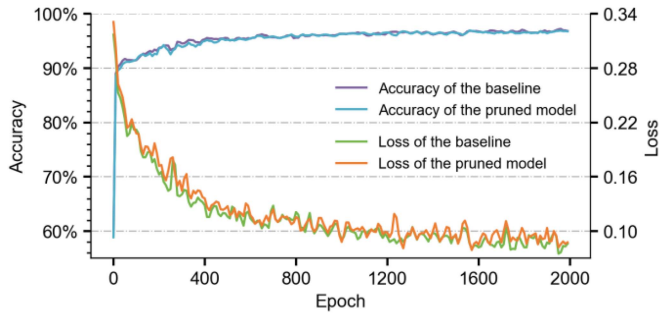


Fig. 3. The change of classification accuracy and loss with the change of epoch for the baseline and the pruned patient-independent model.

TABLE II
COMPARISON BETWEEN THE BASELINE AND THE PRUNED
PATIENT-INDEPENDENT MODEL (EPOCH = 2000)

Model	Accuracy	Sensitivity	Specificity	Parameters (pruning ratio)	MACs (pruning ratio)
Baseline	97.19%	96.44%	97.20%	286.2k	8.2M
Pruned Model	97.09%	96.49%	97.09%	198.3k (30.7% ↓)	6.2M (24.1% ↓)

baseline. The comparison between the baseline and the pruned patient-independent model is summarized in Table II, when the epoch is 2000. The proposed model achieves an average accuracy of 97.19%, an average sensitivity of 96.44%, and an average specificity of 97.20%, which illustrates the effectiveness of the proposed method in seizure detection. Compared with the baseline, the pruned model shows parameter reduction by 30.7%, and decrease of computation (in MACs) by 24.1%, while accuracy drops only by 0.1%.

The results of the patient-specific model (with the proposed kernel-wise pruning) are listed in Table III. For the patient-specific model, the average pruning rate of parameters is 60.2%, while the average pruning rate of MACs is 55.1%. The pruning rate of the compression algorithm on the patient-specific model is much higher than that of the patient-independent model. Considering the fact that the patient-independent model is trained for all subjects while the patient-specific model is only trained for each subject, to obtain an effective patient-independent model is more difficult and time-consuming than to obtain an effective patient-specific model. The results show that our pruning algorithm can effectively prune the redundant convolution kernel in the model according to the difficulty of the task.

The proposed model is also compared with other state-of-the-art methods. Only methods evaluated in the same dataset are included. The comparisons between our proposed methods and other methods are summarized in Table IV. The performance as well as the number of parameters and computations of various models is compared.

Zhang et al. [46] calculate the spectral energy of bandpass filtered EEG sample as features, which are fed into an SVM classifier for seizure detection. The patient-specific method in [46] achieves a good performance, which is somehow only evaluated on a subset of the CHB-MIT dataset. However, even to obtain these relatively simple features, the amount of calculation still reaches 3M MACs.

Zhang et al. [47], Ke et al. [48], and Tian et al. [49] combine feature extraction and deep learning methods. The complex features they use, or the neural networks proposed in those

TABLE III
THE RESULTS OF THE PROPOSED PRUNED PATIENT-SPECIFIC MODEL

Subject	Accuracy	Sensitivity	Specificity	Parameters (pruning ratio)	MACs (pruning ratio)
Chb01	99.9%	100.0%	99.9%	81.3k (71.6% ↓)	2.7M (66.5% ↓)
Chb02	100.0%	100.0%	100.0%	58.9k (79.4% ↓)	2.0M (75.1% ↓)
Chb03	99.9%	97.6%	99.9%	102.9k (64.0% ↓)	3.3M (59.2% ↓)
Chb04	99.9%	100.0%	99.9%	125.0k (56.3% ↓)	4.2M (49.3% ↓)
Chb05	100.0%	100.0%	100.0%	71.8k (74.9% ↓)	2.4M (70.3% ↓)
Chb06	99.9%	100.0%	99.9%	136.3k (52.4% ↓)	4.5M (45.7% ↓)
Chb07	99.9%	93.9%	99.9%	89.5k (68.7% ↓)	3.0M (63.1% ↓)
Chb08	99.7%	91.4%	99.8%	179.0k (37.5% ↓)	5.5M (32.8% ↓)
Chb09	100.0%	96.4%	100.0%	70.5k (75.4% ↓)	2.4M (70.2% ↓)
Chb10	99.9%	100.0%	99.9%	83.9k (70.7% ↓)	2.7M (67.0% ↓)
Chb11	99.9%	100.0%	99.9%	60.7k (78.8% ↓)	2.1M (74.9% ↓)
Chb12	99.1%	93.1%	99.2%	160.5k (43.9% ↓)	5.0M (38.6% ↓)
Chb13	99.2%	91.1%	99.3%	155.8k (45.6% ↓)	5.0M (39.4% ↓)
Chb14	99.6%	100.0%	99.6%	167.4k (41.5% ↓)	5.2M (36.3% ↓)
Chb15	99.7%	97.0%	99.7%	92.0k (67.9% ↓)	3.0M (63.7% ↓)
Chb16	99.6%	100.0%	99.6%	167.6k (41.4% ↓)	5.4M (33.9% ↓)
Chb17	99.8%	96.7%	99.8%	125.4k (56.2% ↓)	4.1M (49.6% ↓)
Chb18	99.7%	93.9%	99.7%	135.2k (52.8% ↓)	4.2M (49.2% ↓)
Chb19	99.9%	100.0%	99.9%	74.2k (74.1% ↓)	2.5M (70.1% ↓)
Chb20	99.8%	96.8%	99.9%	103.4k (63.9% ↓)	3.4M (58.0% ↓)
Chb21	99.9%	95.2%	99.9%	157.4k (45.0% ↓)	4.9M (40.2% ↓)
Chb22	100.0%	100.0%	100.0%	60.0k (79.0% ↓)	2.0M (75.4% ↓)
Chb23	99.7%	97.7%	99.7%	77.1k (73.1% ↓)	2.6M (68.3% ↓)
Chb24	99.5%	89.6%	99.6%	196.0k (31.5% ↓)	6.1M (25.9% ↓)
Average	99.8%	97.1%	99.8%	113.8k (60.2% ↓)	3.7M (55.1% ↓)

works, introduce a huge amount of computation. Specifically, the feature extraction process in [48] accounts for the majority of the computation, which exceeds 1G MACs. The neural networks in [47] and [49] contain 19.8G and 1.9G MACs, respectively. Furthermore, these methods are time-consuming and rely on human expertise.

Abdelhameed et al. [50] apply deep convolutional autoencoder (DCAE) to automatically learn features from raw EEG signals. The features are fed into a Bi-LSTM to perform the classification task, which introduces 83.1M MACs. Furthermore, these computations cannot be parallelized.

Wang et al. [51] and Li et al. [52] adopt 1D CNN to detect seizure from the raw EEG signal. The models in [51] and [52] only contain 106.8k and 72.6k parameters, respectively. The amount of calculations is 10.1M and 20.2M MACs in [51] and [52], respectively. However, their models only contain three convolution layers with short convolution kernels (i.e., 1×3 , 1×5 , 1×8). These two models are shallow and the kernels have narrow receptive field. The feature extraction ability of their patient-specific models is not strong. The sensitivity of their model is only 88.14% and 95.42%, respectively.

The proposed patient-specific model achieves 99.77% accuracy, 97.11% sensitivity, and 99.78% specificity, with only 113.8k parameters and 3.7 M MACs. The amount of computation of the proposed LSN model is even close to the simplest feature extraction method [46]. Compared with other patient-specific models [49], [51], [52], the proposed LSN model

TABLE IV
COMPARISONS TO THE STATE OF THE ART

Study	Method	Evaluation	Accuracy	Sensitivity	Specificity	Parameters	MACs
[46]	SE+SVM	LOOCV Patient-specific	-	97.8%	99.5%	-	3M
[47]	STFT+CNN	5-F-CV Patient-independent	98.26%	-	-	143.7M	19.8G
[48]	Global MIC+2D-CNN	5-F-CV Patient-independent	98.13%	98.85%	97.47%	50k [†]	>1G*
[49]	FFT, WPD+ 2D-CNN, 3D-CNN	5-F-CV Patient-specific	98.33%	96.66%	99.14%	28.8M [‡]	1.9G [‡]
[50]	Raw data+ DCAE+Bi-LSTM	10-F-CV Patient-independent	98.79%	98.72%	98.86%	139.6k	83.1M
[51]	Raw data+ 1D-CNN	LOOCV Patient-specific	99.54%	88.14%	99.62%	106.8k	10.1M
[52]	Raw data+ 1D-CNN+NLSTM	10-F-CV Patient-specific	95.29%	95.42%	95.29%	72.6k	20.2M
This Work (LightSeizureNet)	Raw data+1D-CNN	10-F-CV Patient-independent	97.09%	96.49%	97.09%	198.3k	6.2M
		10-F-CV Patient-specific	99.77%	97.11%	99.78%	113.8k	3.7M

LOOCV: leave-one-out cross validation; 10-F-CV: 10-fold cross validation; SE: spectral energy; SVM: support-vector-machine; STFT: short-time Fourier transform; MIC: maximal information coefficient; FFT: fast Fourier transform; WPD: wavelet packet decomposition; NLSTM: nested Long Short-Term Memory; DCAE: deep convolutional autoencoder.

*: Only feature extraction is counted.

†: Only parameters in the lightweight VGGNet is counted.

‡: Only 3D-CNN is counted.

achieves better classification performance with less computation. Specifically, compared to [49], [51], and [52], the proposed LSN model reduces the amount of calculation by $513.5 \times$, $2.7 \times$, and $5.5 \times$, while enhancing the classification accuracy by 1.4%, 0.2%, and 4.4%, respectively.

The proposed patient-independent model achieves 97.09% accuracy, 96.49% sensitivity, and 97.09% specificity, with only 198.3k parameters and 6.2M MACs. Compared with other patient-independent model with complex feature extraction methods [50] and deep neural network models [47], [48], the proposed LSN model has significant advantages in computational complexity. Meanwhile, the proposed model is also competitive in terms of accuracy, sensitivity, and specificity. Specifically, compared to [50], [47], and [48], the proposed LSN model reduces the amount of calculation by 92.5%, 99.9%, and 99.4%, only with accuracy drop of 1.7%, 1.2%, and 1.0%, respectively. Therefore, the proposed lightweight seizure detection neural network model LSN has the advantages of less parameters and low computation complexity while maintaining competitive accuracy.

E. Interpretability Analysis

The interpretability of neural network is important for the applications in the medical field. Specific fine-grained information, such as the activated brain region and the frequency of brainwave during seizure, can be used for the diagnosis of epilepsy and surgical plan. Benefited from the concise and efficient design, the proposed LSN model can be interpreted from two aspects: EEG channels and kernels. The proposed model can automatically pay different attention to variable EEG channels. The information on EEG channels indicates the location of epileptic wave. Furthermore, due to the kernel-wise pruning mechanism, the model can also capture the importance of various kernels. By analyzing

the frequency band related to the kernels, we can further obtain the information about the type of epilepsy. In this section, the interpretability of the proposed LSN model is presented in detail.

1) Identification of the Activated Brain Region

As described in Section III-C, the fully-connected layer is replaced by the global average pooling structure, which is put before the final output layer. To obtain the response of neural network to each EEG channel, we propose a method based on *EEG Channel Activation Map* (ECAM). With ECAM, we can further identify the activated brain region during seizures.

In the proposed LSN model, global average pooling is applied on time step. GAP outputs the temporal average of the feature map of each EEG channel in the last convolution layer. A weighted sum of these values is used to generate the final output. In [38], the class activation maps are used to localize the discriminative image regions used for image classification. However, the specific time when epileptic waves appear in the EEG sample is not critical because epileptic waves appear repeatedly in the EEG sample (the frequency of ictal pattern is generally greater than 2 Hz [39] and the EEG samples in our dataset last for 2 seconds). A single epileptic wave may be noise or epileptiform discharge between seizures. Averaging the values in the whole EEG sample make the results more robust. Therefore, we compute a weighted sum of all feature maps from each channel to obtain the EEG Channel Activation Maps. Since the proposed LSN model only uses intra-channel convolution operations, all channels are separated. Therefore, the summation of feature maps of each channel represents the response activation of the channel.

For a given EEG sample, let $h_c^k(t)$ represent the activation of kernel k in the c -th EEG channel in the last convolution layer, t denotes the time-step. The result of global average

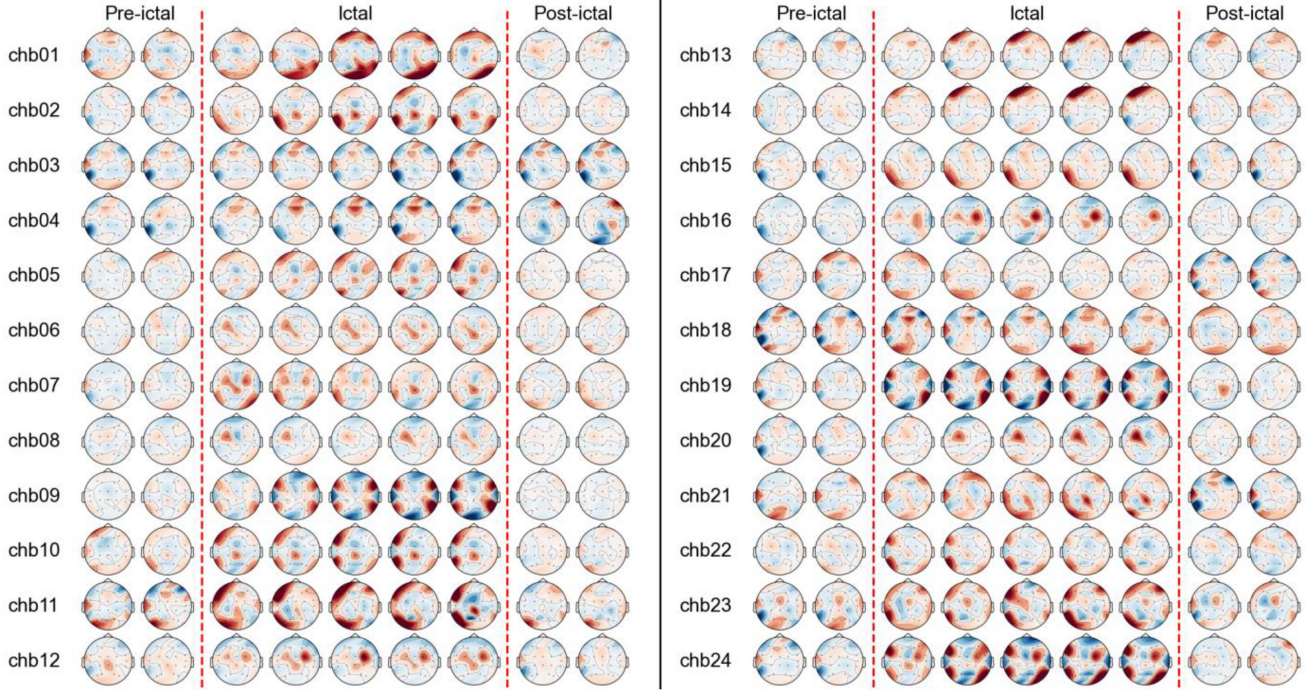


Fig. 4. The EEG Channel Activation Maps (ECAMs) of all subjects obtained from the patient-independent LSN mode.

pooling is derived as $H_c^k = \sum_t h_c^k(t)$. Therefore, for a certain class s (seizure state), we obtain y_s by computing the weighted summation of the output of global average pooling. y_s can be formulated as

$$y_s = \sum_c \sum_k w_c^{k,s} H_c^k = \sum_c A_c^s, \quad (7)$$

where $w_c^{k,s}$ denotes the weight corresponding to class s for kernel k in the c -th EEG channel. A_c^s denotes the weighted sum of all the feature maps of the c -th EEG channel, $\sum_k w_c^{k,s} H_c^k$. y_s is the summation of A_c^s . For a higher A_c^s , there is a higher y_s . Considering there is softmax function at the end of the model, the probability of class s , P_s is $\frac{\exp(y_s)}{\sum \exp(y_s)}$. A higher A_c^s leads to a higher probability of class s . Therefore, A_c^s indicates the importance of the feature maps in the c -th EEG channel for class s . All A_c^s forms the ECAMs A^s .

We plot the ECAMs A^s of all subjects obtained from a trained general model in Fig. 4. The ECAMs are obtained from EEG segments which are 2 seconds before seizure, 5 seconds after the onset of the seizure, and 2 seconds after the seizure. The red color represents larger activation while blue color indicates smaller activation. A large value indicates that the EEG channel has a high probability of epileptic waves. Namely, the model infers a high probability of seizure onset. Combined with the activation information from all EEG channels, the model can accurately judge the seizure status. To show that the ECAMs obtained from the proposed LSN model can locate seizure onset zone, more examples of EEG recordings and the corresponding ECAMs are available at [62].

Currently, physicians manually locate epileptic foci through clinical procedure. They need to find the exact region of the brain causing the epileptic seizures for surgical plan [53], [54]. Repetitive seizures in focal epilepsy tend to have a consistent

pattern of onset and evolution both clinically and electrographically [39]. It is consistent with what we have observed via ECAMs. As shown in Fig 4, the ECAMs vary among subjects and can dynamically display the process of epileptic seizure. For example, the seizures occurred in chb09, chb19, and chb24 are widely distributed. While the origin of seizures in chb13, chb14, chb15, and chb20 are limited to one hemisphere, which are consistent with the definition of focal epilepsy. In short, ECAMs give us the ability to analyze the epileptic foci and judge the type of seizures from the perspective of neural network model. It is promising that the ECAMs obtained from the proposed model can provide fine-grained information for epileptic seizure diagnosis.

2) Identification of the Frequency Band

By employing kernel-wise pruning described in Section III-C, for a well-trained and effective model, the learned weights on kernels can be regarded as the importance of the kernels. When we obtain the importance of EEG channels by ECAMs, we can further analyze the importance of convolution kernel of the EEG channel.

We conduct experiments to obtain more fine-grained information, such as the frequency of brainwave during seizure, for each subject. Take chb14 as an example, we train a patient-specific model for chb14. The model achieves a 99.6% accuracy, 41.5% compression rate of parameters, and 36.3% compression rate of computations. The compression ratio of the patient-specific model is higher than that of the patient-independent model, which suggests that the patient-independent model needs more convolution kernels to increase the feature extraction ability for various subjects. Meanwhile, the patient-specific model is trained for each subject. Only the most important convolution kernels are retained. We can obtain more targeted fine-grained information by analyzing these

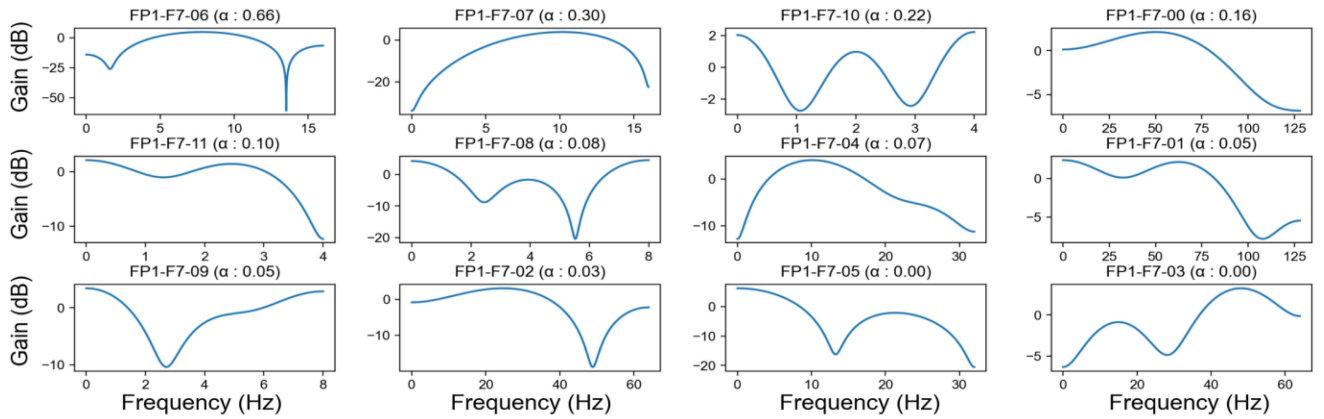


Fig. 5. The frequency response of the convolution kernels of the patient-specific LSN model for chb14.

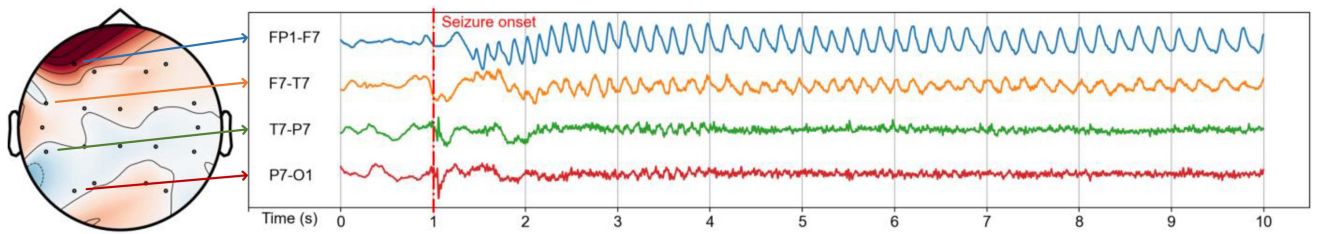


Fig. 6. The example of EEG recording of chb14 during seizure.

most important convolution kernels of the patient-specific model.

Through ECAMs, we obtain the EEG channel “FP1-F7” (located over left frontal lobe) that is most concerned by the patient-specific LSN model for chb14. Then, we rank convolution kernels of EEG channel “FP1-F7” of the first layer according to their importance (i.e., learnt weights α). We analyze the concerned frequency band by calculating the frequency response of each kernel as shown in Fig. 5. From the figure, we can see that the kernel “FP1-F7-6” and “FP1-F7-7” have significantly greater weight than the other kernels, which mean that they are the most important kernels. The frequency response of these two most important kernels “FP1-F7-6” and “FP1-F7-7” is similar to the band-pass filter with pass band at 3.8–11.7 Hz and 4.9–14.3 Hz, respectively. These two frequency bands of EEG signals are reserved by the band-pass filter, which is the frequency band that is most concerned by the patient-specific LSN model. The recordings of the EEG channels around “FP1-F7” of chb14 is shown in Fig. 6. As shown in Fig. 6, an epileptiform discharge occurs on the channel “FP1-F7”, the frequency of which gradually decreases from 6.7 Hz to 4.2 Hz. The frequency band of the epileptiform discharge is consistent with the most concerned frequency band of the convolution kernel as we analyze above. Therefore, the proposed LSN model can automatically extract features from the frequency bands of interest from the raw EEG data and judge the status of epileptic seizure through these features.

V. DISCUSSION

In this work, we propose a lightweight and interpretable neural network model for seizure detection, the performance of which

is competitive compared to the state of the art in terms of accuracy and complexity. The proposed model can be used for wearable seizure detection. The application scenarios include: (a) Labeling seizures in long-term EEG recordings to reduce the workload of EEG annotators; (b) Detecting seizures at the initial stage to send warnings to caregivers, or being implemented in a closed-loop implantable stimulation device to provide responsive electrical stimulation for ambulatory epilepsy treatment. Different application scenarios demand different classification accuracy of the model. For visual inspection of long-term EEG recordings, the objective of seizure detection is to replace the need of reviewing the complete recordings by neurologists. The proposed patient-independent model achieves 97.09% accuracy, 96.49% sensitivity, and 97.09% specificity. The 96.49% sensitivity is sufficiently high to capture most of seizure events. Since seizure generally lasts from a few seconds to a few minutes or even longer, the model with 97.09% specificity can achieve a low false alarm rate for seizure events by majority voting of the results of multiple EEG segments.

The alarming system and the closed-loop implantable stimulation devices require prompt and accurate response for both the onset and termination of seizures. To this end, our proposed models use short EEG segments (e.g., two seconds) and provide result in each EEG segment. In this case, specificity even above 99% can still come along with unacceptable number of false detections. The false positive rate of our proposed patient-specific model is 7.51 false positive per hour, with 99.78% specificity. Despite the existence of noise, such as ocular artifacts (EOG) and muscle activity (EMG), interictal epileptiform discharges (IEDs) are the main cause of false alarms for the proposed models. IEDs consisting of spike or sharp wave discharges are commonly found in patients with epilepsy [61]. For the

application scenario of the alarming system and closed-loop stimulation equipment, post-processing of the classification results could be leveraged to further reduce the false alarms caused by IEDs. For example, we can increase the threshold of positive classification or increase the positive alarm interval. The setting could be customized by the clinical experts for each patient.

VI. CONCLUSION

In this work, we propose a lightweight deep learning model, LightSeizureNet (LSN), for real-time epileptic seizure detection running on ultra-low power wearable devices in non-hospital environment. The proposed LSN model can directly extract high dimensional features from raw EEG signals and classify the seizure states of the samples. The LSN model includes a patient-independent and a patient-specific version, both of which avoid manual feature extractions and high computation complexity, while maintaining good classification accuracy. Dilated one-dimensional (1D) convolution, global average pooling, and kernel-wise pruning are adopted to compress the LSN model. The proposed models are evaluated on the CHB-MIT scalp EEG database. The patient-independent LSN model achieves 97.09% accuracy with 6.2M MACs, while the patient-specific LSN model achieves 99.77% accuracy with 3.7 M MACs, which are competitive compared to the state of the art in terms of accuracy and complexity.

Furthermore, the proposed model is highly interpretable, which is critical for the medical field yet missing in many previous works. We explain the feature extraction ability of the proposed model from two aspects: EEG channels and kernels. By using a uniform approach to explore the interpretability of the proposed model, fine-grained information such as the activated brain region and the frequency of brainwave during seizures is obtained for clinical diagnosis.

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